

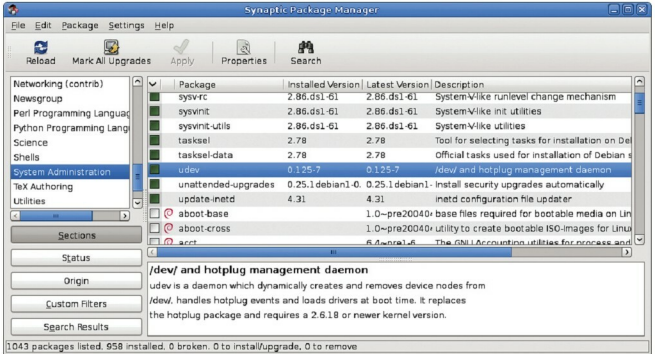


Figure 4: Synaptic package manager

user-friendly, I am not going to get into the details. Most Linux distributions have this GUI package manager.

Debian based tools

apt: apt is a set of core tools found inside the Debian operating system. It provides utilities for the installation and removal of software packages and dependencies on a system.

Installing and managing individual packages involve the following operations:

- Upgradation of packages
- Applying security patch(es)
- Keeping Debian systems up-to-date
- Downloading source .deb files
- Having many GUI and other utilities as front-ends

dpkg: dpkg is a sort of offline installer of Debian packages. It is used to install/manage individual packages.

Syntax	Description	Example(s)
<code>apt install {package}</code>	Installs a new package. If a package is installed, then try to upgrade to the latest version.	<code>apt-get install zip</code> <code>apt-get install lsof samba</code> <code>mysql-client</code>
<code>apt remove {package}</code>	Removes/deletes an installed package except configuration files.	<code>apt-get remove zip</code>
<code>apt --purge remove {package}</code>	Removes/deletes everything including configuration files.	<code>apt-get --purge remove</code> <code>mysql-server</code>
<code>apt update</code> <code>apt upgrade</code>	Resynchronises the package index files and upgrades the Debian Linux system, including security update (Internet access required).	<code>apt-get update</code> <code>apt-get upgrade</code>
<code>apt update</code> <code>apt dist-upgrade</code>	Usually used to upgrade Debian distributions; for example, Woody to Sarge upgrade. 'dist-upgrade', in addition to performing the function of an upgrade, also intelligently handles changing dependencies with new versions of packages; apt-get has a 'smart' conflict resolution system, and it will attempt to upgrade the most important packages at the expense of the less important ones, if necessary.	<code>apt-get update</code> <code>apt-get dist-upgrade</code>
<code>apt-get autoremove</code>	Removes the orphan packages.	<code>apt-get autoremove</code>
<code>apt search</code>	Searches for packages.	<code>apt search lynx</code>

Table 2: Syntax, description and examples of apt